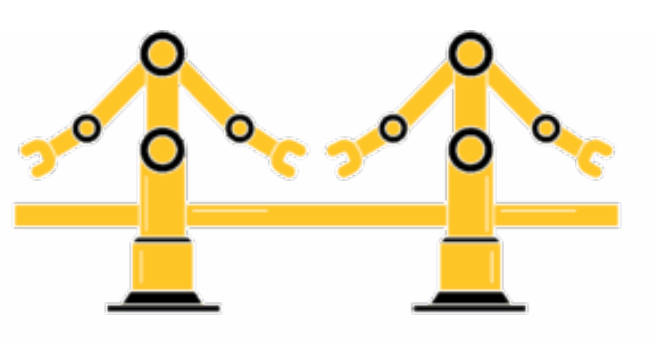




Model-Based Visual Tactile Sensing and Perception System for Compliant Robotic Grippers

Kaiwen Zuo^{1*}, Shuyuan Yang¹, Zonghe Chua¹

¹Case Western Reserve University, Cleveland, OH



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Introduction

Tactile sensing for compliant robotic manipulators involves trade-offs among cost, complexity, robustness, and performance¹.

With the growing integration of RGB-D wrist cameras into robotic systems for control purposes, we propose a **model-based visual tactile sensing and perception framework** for both grasp and manipulation forces from observed deformation.

Method

Hardware Setup

Two Fin Ray-inspired soft gripper jaws, 3D-printed in TPU 95A, were mounted on an electromechanical gripper base and observed using a wrist-mounted Intel RealSense D405 RGB-D camera (Fig. 1(A)).

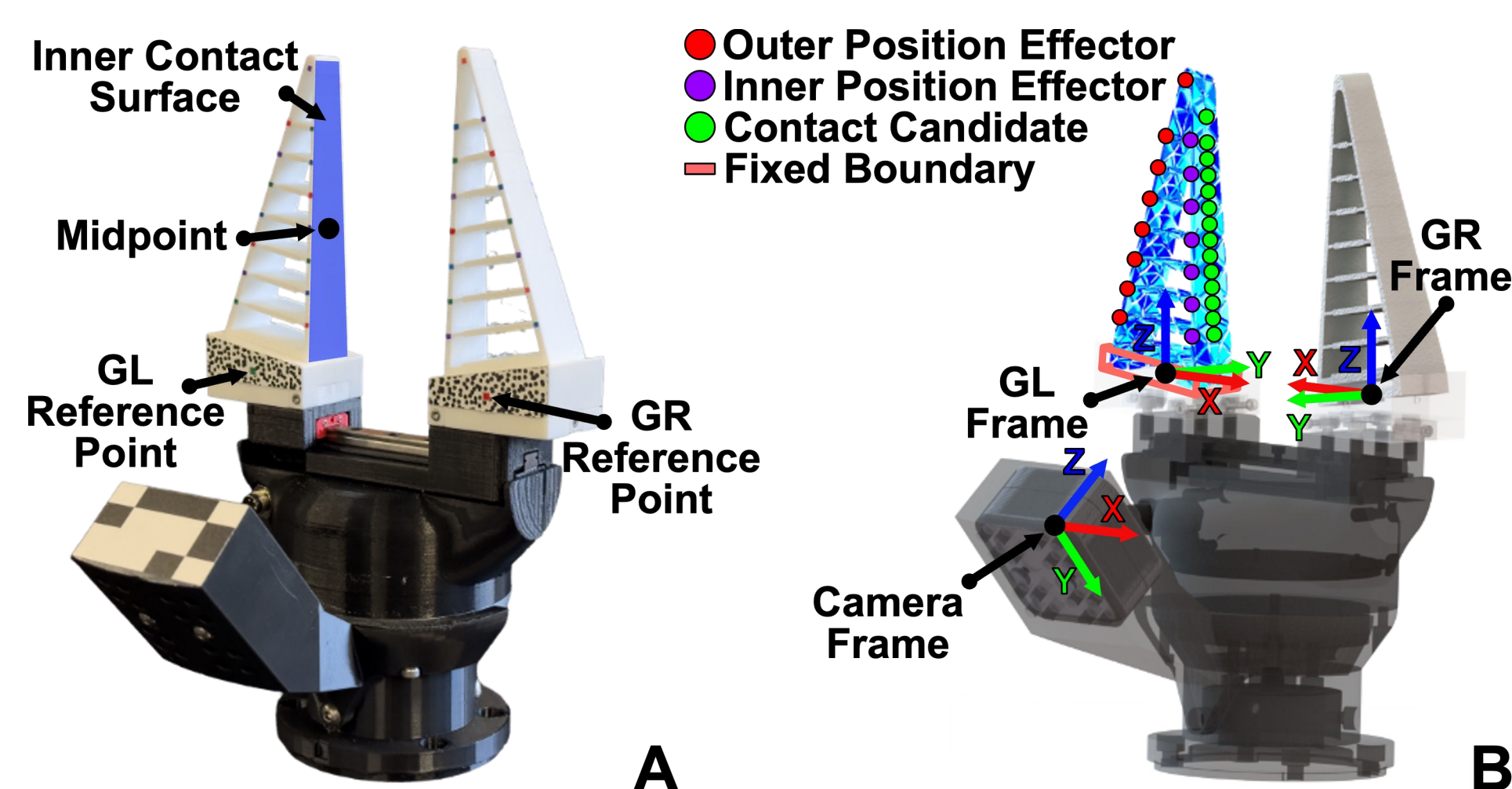


Fig. 1. Physical dual-jaw gripper and its digital twin. (A) Hardware setup. (B) SOFA simulation setup.

Simulation Setup

The iFEA simulation is implemented in SOFA with position-effector and actuator constraints. A QP-based inverse solver estimates the forces applied at the estimated contact positions by minimizing the error between the simulated position effectors and their RGB-D-observed targets.

System Framework

RGB-D signals are processed to estimate contact information and gripper keypoints, which are then fed into iFEA for per-jaw force estimation.

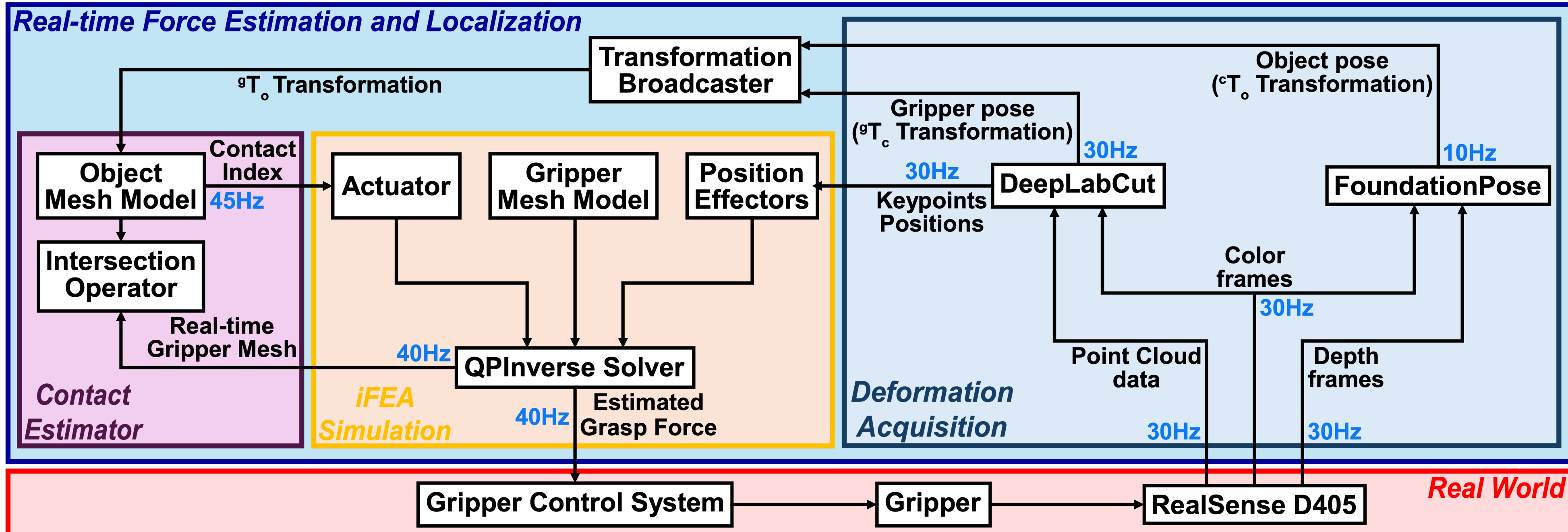


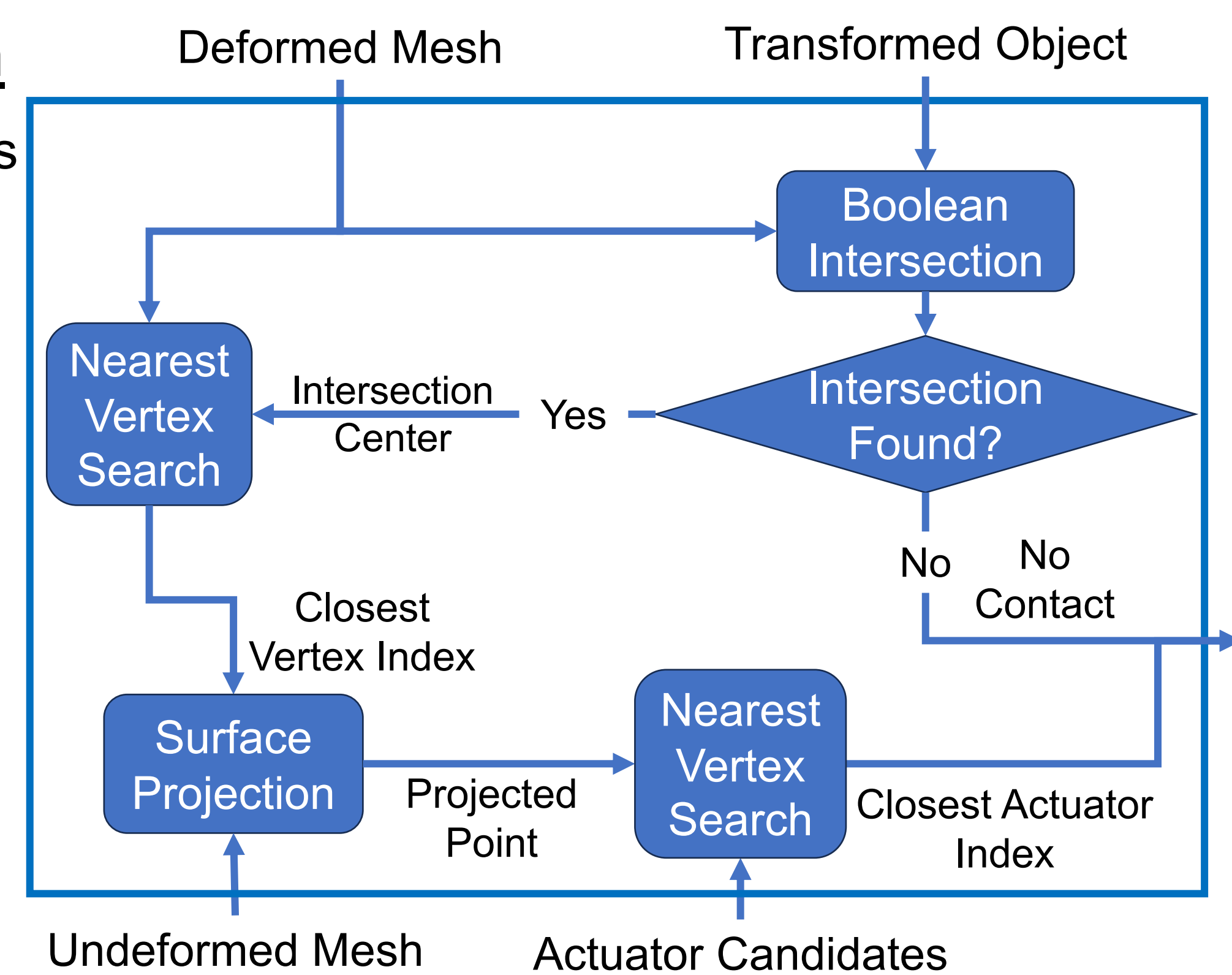
Fig. 2. Flow chart for the tactile sensing and perception system pipeline

Key Points Acquisition and Calibration

DeepLabCut (DLC) tracks 15 structural key points and one reference point per gripper jaw. The structural key points correspond to the position effectors in simulation, while the reference point tracks the gripper poses. A confidence threshold, followed by a spatial bounding box filter, rejects misidentified key points.

Model-based Contact Estimator*

The proposed algorithm (Algorithm 1) iteratively updates contact locations, enabling dynamic estimation during manipulation without relying on a fixed initialization. To improve generalizability across objects, we introduce an image-based mesh reconstruction approach that provides the required 3D object geometry.



Algorithm 1: Iterative Contact Localization Algorithm.

Static Force Evaluation

- We performed a static force evaluation by applying five different load levels to the gripper while interacting with three cylinders of varying diameters: 15, 25, and 35 mm. Each cylinder was tested at three distinct contact positions: upper, middle, and lower (Figs. 3(B)-(D)).

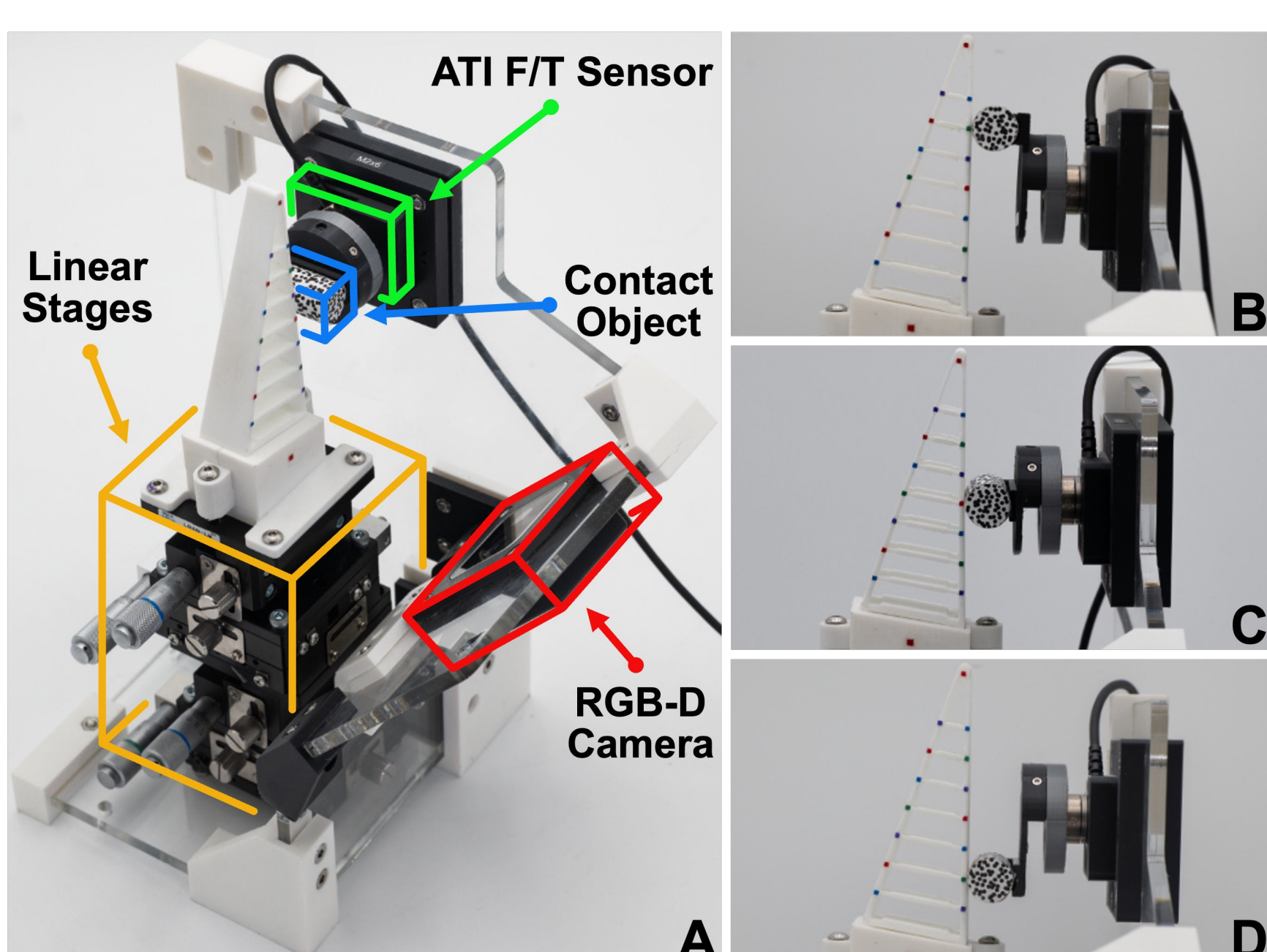


Fig. 3. Benchtop setup for static evaluation and configurations of contact positions. (A) Experimental setup. (B–D) The gripper is set to contact a 15 mm diameter cylinder at (B) upper, (C) middle, and (D) lower positions.

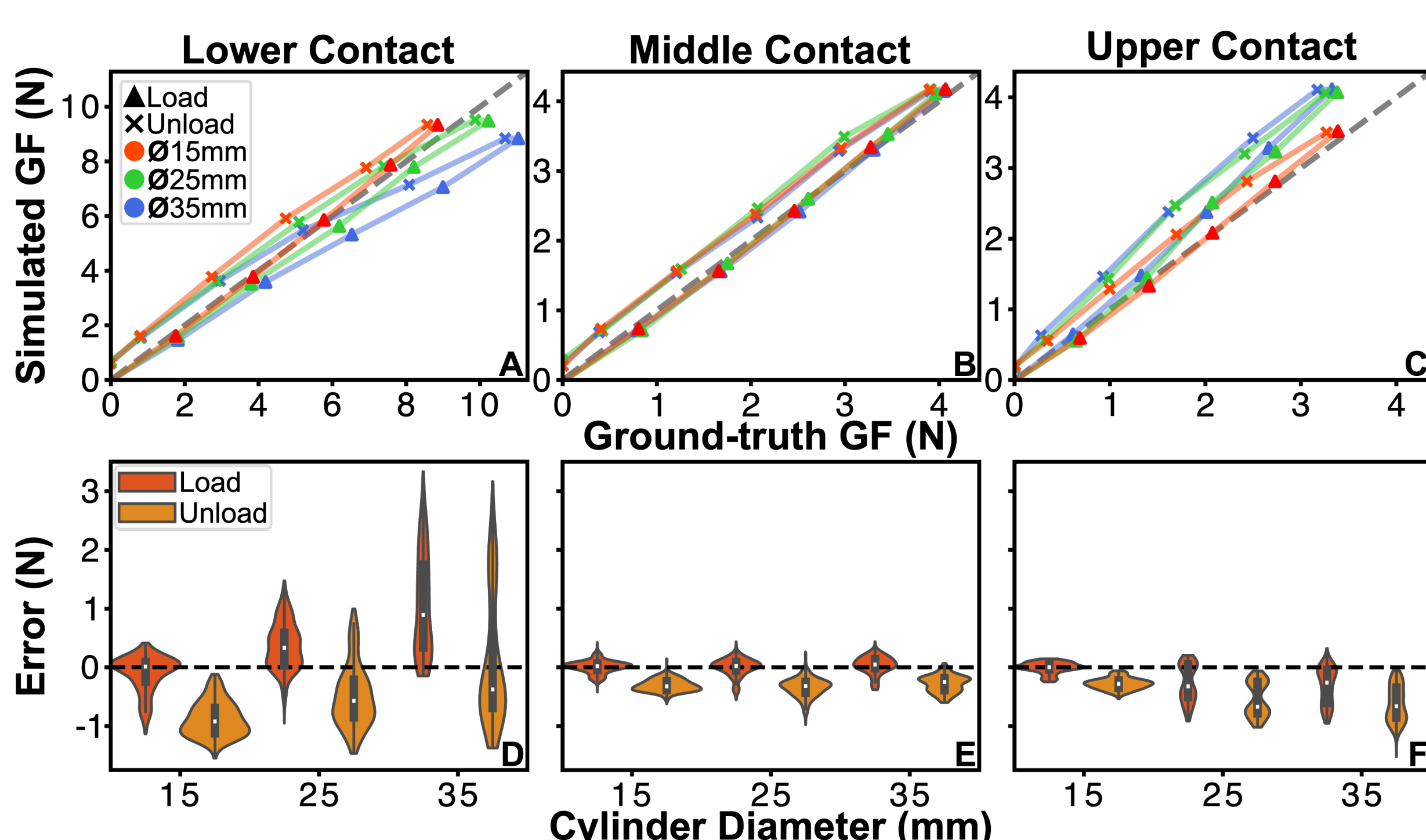


Fig. 4. Effects of contact position and object size on grasp-force estimation. (A–C) F_{sim} vs F_{gt} during loading and unloading for 15, 25, and 35 mm cylinders at (A) lower, (B) middle, and (C) upper contact positions. (D–F) Corresponding error distributions for (D) lower, (E) middle, and (F) upper positions.

On-robot Grasp and Manipulation Force Evaluation

- For grasp-force evaluation, the dual-jaw gripper was mounted on a Kinova robotic arm to grasp three differently shaped objects instrumented with load cells.
- For manipulation-force and lighting sensitivity evaluation, the gripper grasped a calibration weight while rotating 360° counterclockwise and back under conditions with and without external lighting.
- Finally, force-controlled potato-chip grasping was performed.

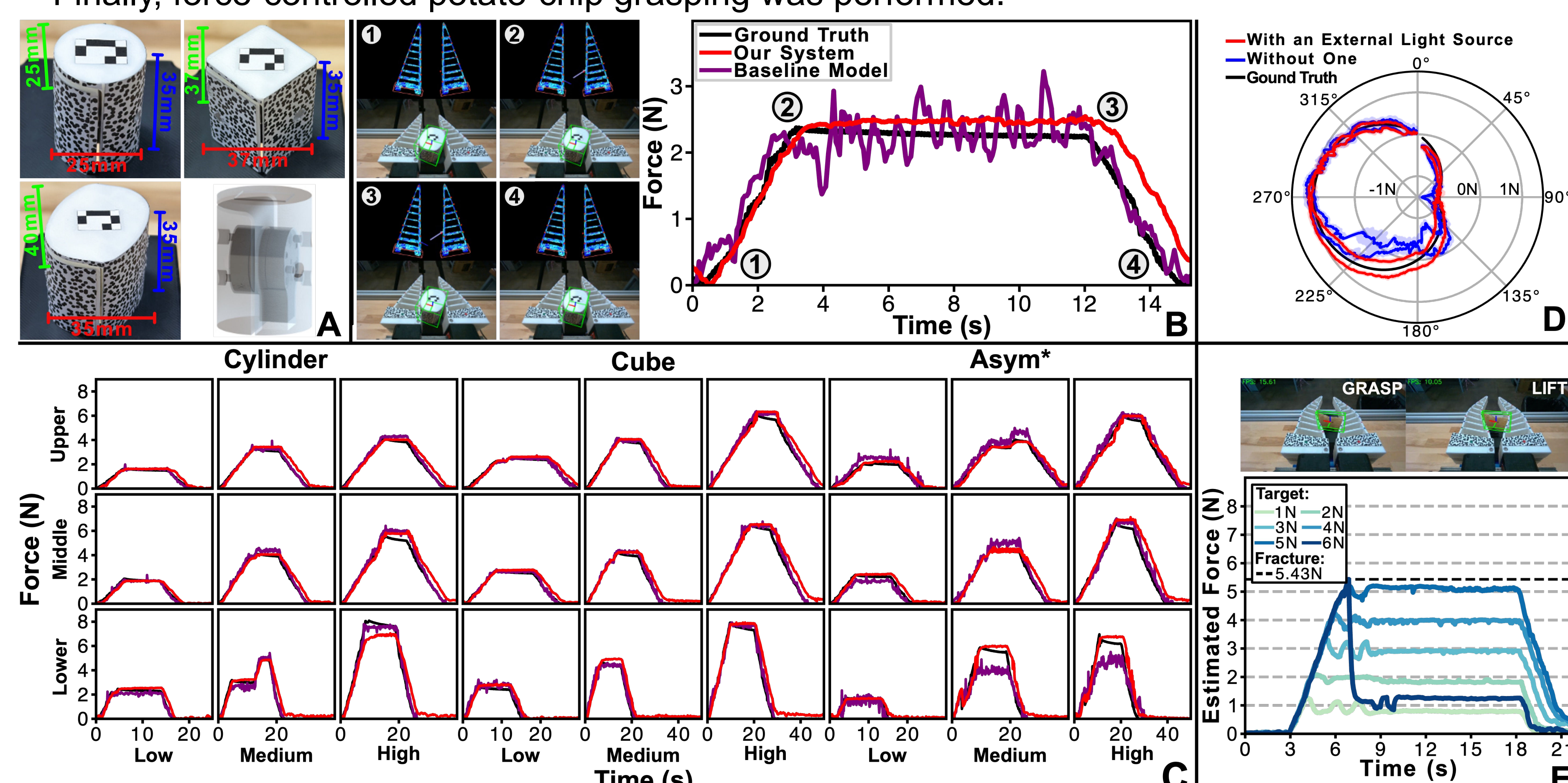


Fig. 5. On-robot evaluation. (A) Load-cell-integrated test objects and CAD designs. (B) Force estimation for the asymmetric object under a representative grasping condition. (C) Grasp-force estimation across objects and conditions; **Asym*** denotes an unseen asymmetric object. (D) Force trajectory during gripper rotation. (E) Force-controlled potato-chip grasping and lifting.

Conclusion

- We developed a model-based visual force sensing framework combining RGB-D perception with real-time iFEA was developed for compliant grippers.
- The system enables reliable force estimation on unseen objects under occlusion, outperforming an end-to-end learning baseline.
- Force-controlled potato-chip grasp demonstrates its potential for delicate robotic manipulation.

References

- J. Qu, B. Mao, Z. Li, Y. Xu, K. Zhou, X. Cao, Q. Fan, M. Xu, B. Liang, H. Liu, X. Wang, X. Wang, *Recent Progress in Advanced Tactile Sensing Technologies for Soft Grippers*. *Adv. Funct. Mater.* 2023, 33, 2306249.